

INTERNSHIP PROJECT REPORT

on

Mutual Fund Performance & Industry Analytics

An End-to-End Data Analytics Project using Python, PostgreSQL, and Power BI

Submitted in partial fulfilment of the requirements of the

Data Analyst Internship Programme

Organization	Bluestock Fintech Pvt. Ltd.
Intern Name	Saurabh Soni
Domain	Financial Data Analytics
Internship Duration	23 June 2026 – 26 July 2026
Report Date	01 August 2026

Certificate

This is to certify that Saurabh Soni has successfully completed the internship project titled “Mutual Fund Performance & Industry Analytics” at Bluestock Fintech Pvt. Ltd. during the period 23 June 2026 to 26 July 2026, under the Data Analyst Internship Programme.

During this period, the intern was engaged in the collection, cleaning, and structuring of mutual fund industry data; the design and population of a relational database in PostgreSQL; the development of SQL-based business queries; exploratory data analysis and performance analytics using Python; and the design of interactive Power BI dashboards for mutual fund performance and investor behaviour analysis.

The intern demonstrated a sound understanding of the end-to-end data analytics workflow, from raw data handling to the presentation of actionable business insights, and conducted the assigned work with diligence and professionalism.

We place on record our appreciation for the intern’s contribution and wish them continued success in their professional journey.

Date: 01 August 2026

Place: Bluestock Fintech Pvt. Ltd.

Authorized Signatory
Bluestock Fintech Pvt. Ltd.

Acknowledgement

I would like to express my sincere gratitude to Bluestock Fintech Pvt. Ltd. for providing me with the opportunity to undertake this internship and work on the “Mutual Fund Performance & Industry Analytics” project. This experience has been invaluable in strengthening my practical understanding of data analytics in the financial services domain.

I am thankful to my mentors and the project team for their continuous guidance, constructive feedback, and encouragement throughout the internship period from 23 June 2026 to 26 July 2026. Their support enabled me to work through the complete analytics lifecycle — from data collection and database design to dashboard development and insight generation — with clarity and confidence.

I also acknowledge the open-source tools and technologies, including Python, PostgreSQL, and Power BI, that formed the technical backbone of this project. Finally, I extend my appreciation to everyone who directly or indirectly contributed to the successful completion of this internship project.

Saurabh Soni

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1. Executive Summary

This report documents the design, development, and analytical findings of the “Mutual Fund Performance & Industry Analytics” project, undertaken as part of a Data Analyst internship with Bluestock Fintech Pvt. Ltd. between 23 June 2026 and 26 July 2026. The project covers the Indian mutual fund industry and combines relational database design, SQL-based business analysis, Python-based exploratory and performance analytics, and Power BI dashboard development into a single, end-to-end analytics workflow.

The underlying data was organized into a ten-table PostgreSQL schema covering fund master details, NAV history, AUM by fund house, SIP inflows, category-wise inflows, industry folio growth, scheme-level performance metrics, portfolio holdings, investor transactions, and benchmark index levels. Seventeen SQL queries were written to answer specific business questions such as fund house ranking by AUM, category-wise expense ratios, top-performing schemes, and fund rankings by Sharpe ratio.

Python was used to validate and extend this analysis: daily returns, annualized returns, CAGR, annualized volatility, Sharpe ratio, and Sortino ratio were independently calculated from NAV history and cross-checked against the pre-computed values in the `scheme_performance` table. The results were consistent with the dataset figures, confirming the integrity of the performance metrics used downstream.

Three interactive Power BI dashboards were built on top of this data: a Mutual Fund Performance & Industry Analytics dashboard, a Mutual Fund Risk & Performance Analytics dashboard, and an Investor Behavior & Transaction Analytics dashboard, along with a SIP & Market Trends dashboard. Together these show that the analysed industry sample carries approximately ₹39 million in total AUM across 40 schemes, an average 5-year return of 14.52%, and total SIP inflows of ₹940K in the dataset window, with small-cap and mid-cap equity schemes leading on long-term returns and liquid/short-duration schemes leading on Sharpe and Sortino ratios.

The report concludes with business insights and recommendations for AMCs and investors, along with an honest account of the project’s limitations and scope for future enhancement, such as automated data refresh pipelines and predictive modelling of NAV trends.

2. Company Overview

Bluestock Fintech Pvt. Ltd. is a fintech organization focused on financial-market data, analytics, and investor-education solutions. The company works at the intersection of capital markets and technology, building tools and internship programmes that give aspiring data professionals hands-on exposure to real-world financial datasets and business problems.

As part of its internship initiatives, Bluestock Fintech assigns project-based assignments that mirror the kind of analytical work performed by data teams inside asset management companies (AMCs), brokerages, and fintech platforms. The “Mutual Fund Performance & Industry Analytics” project is one such assignment, designed to give the intern practical experience across the full data analytics stack — database design, SQL, Python, and business intelligence dashboarding — applied specifically to the Indian mutual fund industry.

3. Project Overview

The project simulates the analytics function of a mutual fund research or investment-advisory desk. It uses a structured dataset covering fund master information, NAV history, AUM trends, SIP inflows, category-wise inflows, folio growth, scheme performance ratios, portfolio holdings, investor transaction records, and benchmark index levels for the Indian mutual fund industry.

The workflow followed for the project is summarised below:

- Data Collection — assembling mutual fund industry data across schemes, fund houses, NAV history, SIP flows, and investor transactions.
- Data Cleaning using Python — preparing cleaned CSV datasets for downstream loading and analysis.
- PostgreSQL Database Design — structuring the cleaned data into a normalized, ten-table relational schema.
- SQL Business Analysis — writing seventeen business-focused SQL queries against the PostgreSQL database.
- Exploratory Data Analysis (EDA) — profiling and visualizing the data in Python using pandas, Matplotlib, and Seaborn.
- Power BI Dashboard Development — building interactive dashboards for performance, risk, and investor-behaviour analytics.
- Business Insights — translating analytical results into insights relevant to AMCs and investors.

- Recommendations — proposing data-driven actions based on the analysis.

This workflow reflects an industry-representative analytics pipeline, moving data from raw storage through a relational database, into statistical analysis, and finally into decision-ready dashboards.

4. Problem Statement & Objectives

4.1 Problem Statement

Retail investors, financial advisors, and AMC business teams need a consolidated, structured view of the mutual fund industry to make informed decisions — covering scheme-level returns, risk-adjusted performance, category-wise fund flows, and investor behaviour patterns. In practice, this information is often scattered across NAV feeds, AMC fact sheets, and transaction systems, making it difficult to compare schemes on a like-for-like basis or to identify industry-wide trends quickly. This project addresses that gap by consolidating mutual fund and investor data into a single relational database and presenting it through SQL analysis, Python-based analytics, and interactive Power BI dashboards.

4.2 Objectives

- To design and implement a normalized PostgreSQL database for mutual fund, performance, portfolio, and investor-transaction data.
- To write SQL queries that answer specific business questions on fund houses, categories, returns, and rankings.
- To perform exploratory data analysis in Python to understand NAV trends, AUM growth, SIP inflows, and category-wise inflows.
- To calculate and validate key performance and risk metrics — annualized return, CAGR, volatility, Sharpe ratio, and Sortino ratio — directly from NAV history.
- To design interactive Power BI dashboards covering performance, risk, and investor-behaviour analytics for both AMC stakeholders and retail investors.
- To translate the analysis into actionable business insights and recommendations for fund selection, marketing, and product strategy.

5. Tools & Technologies

The project was executed using a combination of database, programming, analytics, and visualization tools that together represent a typical modern data-analytics stack:

Category	Tool / Technology	Purpose
Database	PostgreSQL	Relational storage and querying of mutual fund and investor data
Programming	Python (Pandas, NumPy)	Data handling, transformation, and metric calculation
Visualization (Python)	Matplotlib, Seaborn	Exploratory data analysis charts and statistical plots
Business Intelligence	Power BI	Interactive dashboards for performance, risk, and behaviour analytics
Query Language	SQL	Business-question-driven data querying
Version Control	GitHub	Source control and project portfolio hosting
Development Environment	Google Colab / Jupyter Notebook	Python-based analysis and visualization

Table 1: Tools and technologies used in the project

6. Dataset Description

The project is built on a set of ten related datasets covering the mutual fund industry, loaded from cleaned CSV files into a PostgreSQL database. Each table captures a distinct aspect of the mutual fund ecosystem, from static fund attributes to time-series performance and investor-level transactions.

Table	Description
fund_master	Master list of mutual fund schemes with fund house, category, plan, launch date, benchmark, expense ratio, and risk category
nav_history	Daily NAV records per scheme, used to derive returns, volatility, and risk-adjusted ratios
aum_by_fund_house	Monthly AUM (in ₹ crore and ₹ lakh crore) reported by each fund house
scheme_performance	Pre-computed scheme-level performance and risk metrics: 1/3/5-year returns, alpha, beta, Sharpe and Sortino ratios, standard deviation, and maximum drawdown
monthly_sip_inflows	Monthly SIP inflow amounts, active and new SIP accounts, SIP AUM, and year-on-year growth
category_inflows	Monthly net inflows by fund category
industry_folio_count	Monthly industry-wide folio counts split by equity, debt, hybrid, and other categories
portfolio_holdings	Stock-level portfolio holdings per scheme with sector, weight, and market value
investor_transactions	Investor-level SIP, lumpsum, and redemption transactions with demographic and geographic attributes
benchmark_indices	Daily closing values of relevant benchmark indices for comparison against scheme performance

Table 2: Dataset tables used in the PostgreSQL database

Across these ten tables, the dataset used for the dashboards represents 40 mutual fund schemes, industry AUM figures aggregated to roughly ₹39 million (in the dashboard's display units) across the top ten fund houses, and investor transaction records spanning multiple states, age groups, and payment modes.

7. ETL Process & Data Cleaning Methodology

7.1 ETL Workflow

Raw mutual fund and investor data was processed through a standard Extract-Transform-Load (ETL) sequence before being loaded into PostgreSQL. Cleaned datasets — for example 01_fund_master.csv, 02_nav_history.csv, 03_aum_by_fund_house.csv, and 04_monthly_sip_inflows.csv, along with corresponding files for category inflows, folio counts,

scheme performance, portfolio holdings, investor transactions, and benchmark indices — were read into Python using Pandas as the transformation layer, before being structured for loading into the ten-table PostgreSQL schema described in Section 6.

7.2 Data Preparation Steps Observed in the Notebooks

The exploratory and performance-analytics notebooks used in this project (03_eda_analysis.ipynb and 04_performance_analytics.ipynb) work exclusively with pre-cleaned CSV files, and the following preparation steps are directly evidenced in that code:

- Date standardisation — date and month columns across nav_history, aum_by_fund_house, monthly_sip_inflows, category_inflows, industry_folio_count, and benchmark_indices were converted to proper datetime types using `pd.to_datetime()` to enable correct time-series sorting and plotting.
- Chronological sorting — nav_history was sorted by amfi_code and date prior to return calculation, ensuring that day-over-day NAV changes were computed in the correct sequence for every scheme.
- Derived-field handling — daily returns were calculated using a percentage-change transformation on NAV; the resulting missing value on each scheme's first trading day (where no prior NAV exists) was identified and treated as an expected structural gap rather than a data-quality error.
- Join-key consistency — the amfi_code field was used consistently as the join key across fund_master, nav_history, and scheme_performance, allowing scheme names and categories to be reliably attached to performance and risk metrics.
- Schema-level integrity — the PostgreSQL schema (schema.sql) enforces primary keys on fund_master and scheme_performance and foreign-key constraints from nav_history, investor_transactions, and portfolio_holdings back to fund_master, which prevents orphaned records from entering the analytical tables.

A dedicated data-cleaning notebook was referenced in the project's documentation but was not among the files supplied for this report; accordingly, this section describes only the cleaning and preparation steps that are directly evidenced in the uploaded schema, SQL scripts, and analysis notebooks, rather than assuming additional steps that cannot be verified.

8. PostgreSQL Database Design

The project data is organized in a PostgreSQL database (bluestock_mf) containing a single public schema with ten tables, as confirmed directly from the project's pgAdmin 4 object explorer shown below.

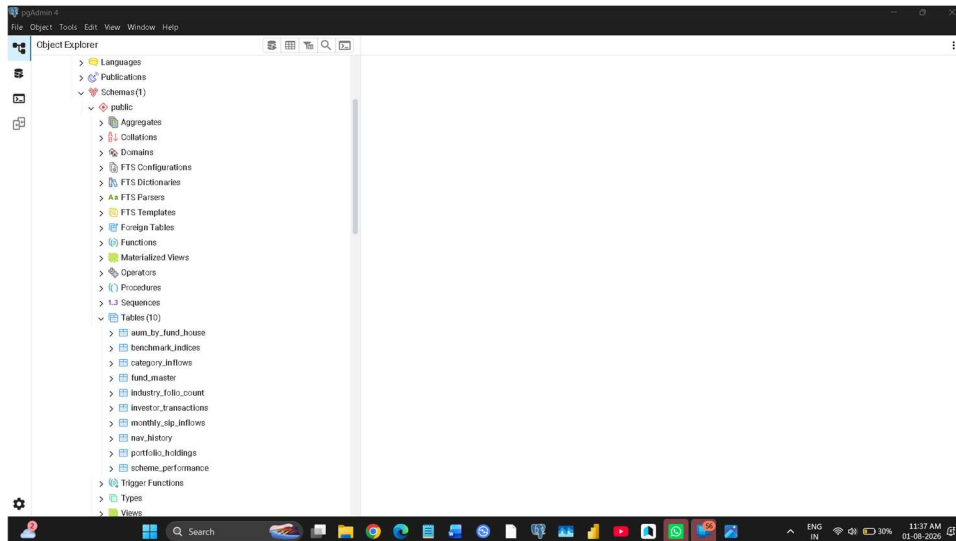


Figure 1: PostgreSQL schema (public) with ten tables, viewed in pgAdmin 4

8.1 Table Relationships

Two tables act as parent (reference) tables in the schema: `fund_master`, keyed on `amfi_code`, and `scheme_performance`, which also carries `amfi_code` as its primary key and a foreign key back to `fund_master`. Three tables — `nav_history`, `investor_transactions`, and `portfolio_holdings` — declare explicit foreign-key constraints on `amfi_code` referencing `fund_master`, ensuring that every NAV record, investor transaction, and portfolio holding is tied to a valid scheme. The remaining tables (`aum_by_fund_house`, `benchmark_indices`, `category_inflows`, `industry_folio_count`, and `monthly_sip_inflows`) are industry-level, time-series tables that are not scheme-specific and are therefore not linked by foreign keys, but share common date/month fields that allow them to be joined analytically with the scheme-level tables where needed.

8.2 Advantages of the Relational Design

- Referential integrity — foreign-key constraints prevent NAV, transaction, or holdings records from referencing a non-existent scheme.
- Query efficiency — the `amfi_code` primary/foreign key structure allows fast, indexed joins between `fund_master`, `scheme_performance`, `nav_history`, and `investor_transactions`.

- Data consistency — storing fund attributes once in fund_master and referencing them elsewhere avoids duplication and update anomalies.
- Analytical flexibility — the separation of static (fund_master), time-series (nav_history, aum_by_fund_house), and transactional (investor_transactions) tables mirrors how real-world AMC data platforms are structured, making the schema directly extensible for further analysis.

9. SQL Business Analysis

Seventeen SQL queries (queries.sql) were written against the PostgreSQL database to answer specific business questions spanning scheme discovery, fund-house ranking, category-level analysis, and performance ranking. The queries progress from simple exploration to window-function-based ranking and multi-table joins. Key queries are summarised below rather than reproduced in full.

#	Business Objective	Business Value
1–2	List all schemes and count total schemes in fund_master	Establishes the size and scope of the fund universe under analysis
3–4	List distinct fund categories; filter schemes with expense ratio below 1%	Identifies low-cost investment options across categories for cost-conscious investors
5	Rank the top 10 fund houses by total AUM	Identifies the industry's dominant asset managers by scale
6–7	Average expense ratio by category; categories with more than 5 schemes	Flags which categories are more expensive on average and which are well-populated with scheme choices
8	Top 10 funds by 5-year return	Surfaces the strongest long-term performers for investor shortlisting
9	Average NAV per scheme from nav_history	Provides a quick reference NAV level per scheme across the historical window
10–12	Monthly SIP inflow trend; transactions by state; category-wise net inflows	Tracks systematic investment momentum and geographic/category demand patterns
13	Rank all funds by Sharpe ratio using RANK() OVER()	Produces an industry-wide risk-adjusted performance leaderboard
14	Top 5 funds per risk grade using ROW_NUMBER() PARTITION BY	Gives risk-appropriate recommendations within each risk bucket
15–16	Total holding weight by sector; peak benchmark index values	Assesses sector concentration in portfolios and benchmark performance ranges

#	Business Objective	Business Value
17	INNER JOIN of fund_master and scheme_performance by 5-year return	Combines scheme identity with performance metrics for consolidated reporting

Table 3: Summary of key SQL business queries and their objectives

Query 5 (top fund houses by AUM) was executed and validated directly in pgAdmin 4 during the project, as shown below, confirming that SBI Mutual Fund, ICICI Prudential MF, and HDFC Mutual Fund are the three largest fund houses by AUM in the dataset.

fund_house	total_aum
SBI Mutual Fund	8491000
ICICI Prudential MF	6293000
HDFC Mutual Fund	5722000
Nippon India MF	3990000
Kotak Mahindra MF	3502000
Aditya Birla Sun Life MF	3077000
UTI Mutual Fund	2681000
Axis Mutual Fund	2521000
Mirae Asset MF	1561000
DSP Mutual Fund	1490000

Figure 2: SQL query and result set for “Top 10 Fund Houses by AUM,” executed in pgAdmin 4

These SQL outputs feed directly into the corresponding Power BI visuals discussed in Section 11, providing a validated, queryable foundation for the dashboards rather than ad-hoc, dashboard-only calculations.

10. Exploratory Data Analysis & Performance Analytics (Python)

Python analysis was carried out across two notebooks: 03_eda_analysis.ipynb, which profiled and visualized the ten cleaned datasets using Pandas, Matplotlib, and Seaborn; and 04_performance_analytics.ipynb, which calculated and independently validated key performance and risk metrics directly from nav_history.

10.1 Key EDA Findings

- NAV & AUM trends: the top mutual funds and leading fund houses showed an overall upward NAV and AUM trend across the 2022–2025 window, with some schemes exhibiting noticeably higher volatility than others.
- SIP inflows and industry folios: both rose consistently from 2022 to 2025, evidencing growing retail investor confidence and expanding market penetration, while Flexi Cap, Large & Mid Cap, and Mid Cap categories led on net inflows.
- Return vs risk: Small Cap funds combined the highest 5-year returns with the highest volatility, while Gilt funds sat at the low-risk, low-return end — a clear illustration of the risk-return trade-off.
- Portfolio & sector concentration: holdings were concentrated in a small number of large-cap names (Power Grid Corporation and HDFC Bank among the highest-weighted), with Banking & Financial Services the largest single sector exposure.
- Transaction behaviour: SIP transactions dominated the transaction-type mix by volume, and average transaction amounts varied modestly by age group, with middle-aged investors contributing somewhat higher average amounts.

10.2 Performance Metrics Calculated & Validated

Using nav_history, daily return was computed as the percentage change in NAV; this was compounded into annualized return and CAGR, and used to derive annualized volatility (standard deviation of daily returns $\times \sqrt{252}$). Sharpe ratio was calculated as (annualized return – 6.5% risk-free rate) / volatility, and Sortino ratio used the same excess return divided by downside deviation (based only on negative daily returns). Both calculated ratios were cross-checked against the pre-computed sharpe_ratio and sortino_ratio fields in scheme_performance and were found to be closely aligned, validating the dataset's risk metrics.

10.3 Key Performance Findings

- Small Cap and Mid Cap schemes dominated the top-10 rankings by 5-year return, annualized return, and CAGR, with ABSL Small Cap Fund – Regular – Growth recording the highest 5-year return among the analysed schemes.
- Liquid and short-duration debt schemes achieved disproportionately high Sharpe and Sortino ratios due to their low volatility, despite modest absolute returns — a different leaderboard from the pure-return ranking.
- A risk-grade comparison of 5-year returns confirmed the expected pattern: High and Very High risk-grade schemes delivered higher median returns with wider variability, while Low-risk schemes showed narrower, lower return distributions.
- No strong linear relationship was found between expense ratio and 5-year return, and correlation analysis showed Sharpe and Sortino strongly related to each other, volatility positively related to Beta, and maximum drawdown negatively related to returns.

This code-verified process means every major KPI shown in the Power BI dashboards in Section 11 can be traced back to a specific SQL query or Python calculation, rather than relying purely on dashboard-level aggregation.

11. Power BI Dashboard Analysis

Four interconnected Power BI dashboard pages were developed from the PostgreSQL data model, each targeting a distinct analytical audience: overall industry and performance monitoring, risk analytics, investor-behaviour analytics, and SIP/market trend tracking. Every KPI card and chart on these pages is traceable to the SQL queries and Python calculations described in Sections 9–10. All four pages use consistent slicers, page navigation, and Bluestock branding for a unified analyst experience.

11.1 Mutual Fund Performance & Industry Analytics Dashboard

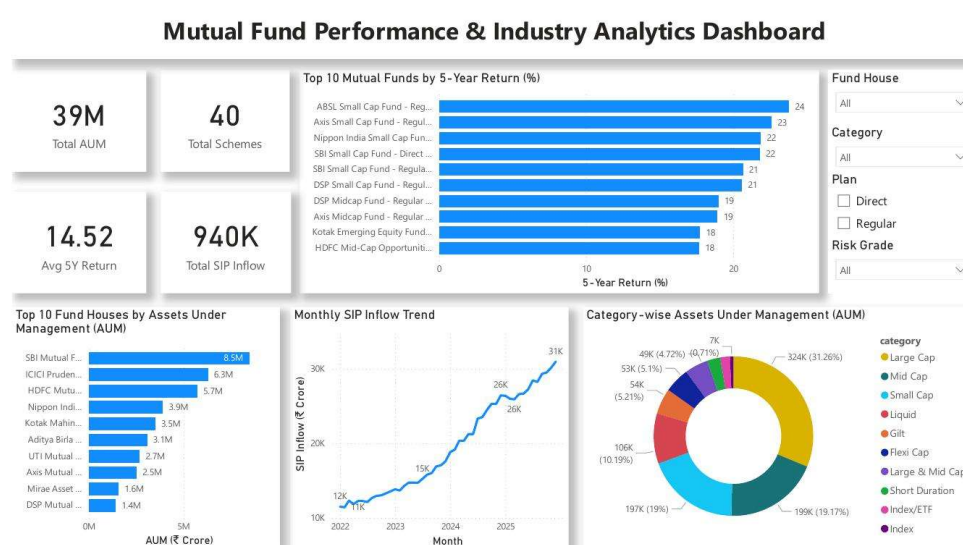


Figure 3: Mutual Fund Performance & Industry Analytics Dashboard — Power BI

This landing page presents four headline KPI cards — Total AUM (₹39M), Average 5-Year Return (14.52%), Total Schemes (40), and Total SIP Inflow (₹940K) — giving a one-glance summary of industry scale and performance.

- Top 10 Mutual Funds by 5-Year Return: a horizontal bar chart led by small-cap and mid-cap equity schemes (ABSL, Axis, Nippon India, and SBI Small Cap Funds returning 21–24%), confirming the EDA finding that equity-oriented categories dominate long-term returns.
- Top 10 Fund Houses by AUM: SBI Mutual Fund (₹8.5M), ICICI Prudential MF (₹6.3M), and HDFC Mutual Fund (₹5.7M) lead the fund-house ranking, matching the SQL query result validated in Figure 2.
- Category-wise AUM: Large Cap (31.26%), Mid Cap (19.17%), and Small Cap (19%) categories together account for close to 70% of AUM, with Liquid, Gilt, Flexi Cap, and index-linked categories making up the remainder.

- Monthly SIP Inflow Trend: a rising line chart from 2022 to 2025, consistent with the steady SIP-growth trend identified in the EDA.

Fund House, Category, Plan (Direct/Regular), and Risk Grade slicers allow the user to interactively filter every visual on this page, making it suitable for both a quick industry snapshot and a focused, category-level review.

11.2 Mutual Fund Risk & Performance Analytics Dashboard

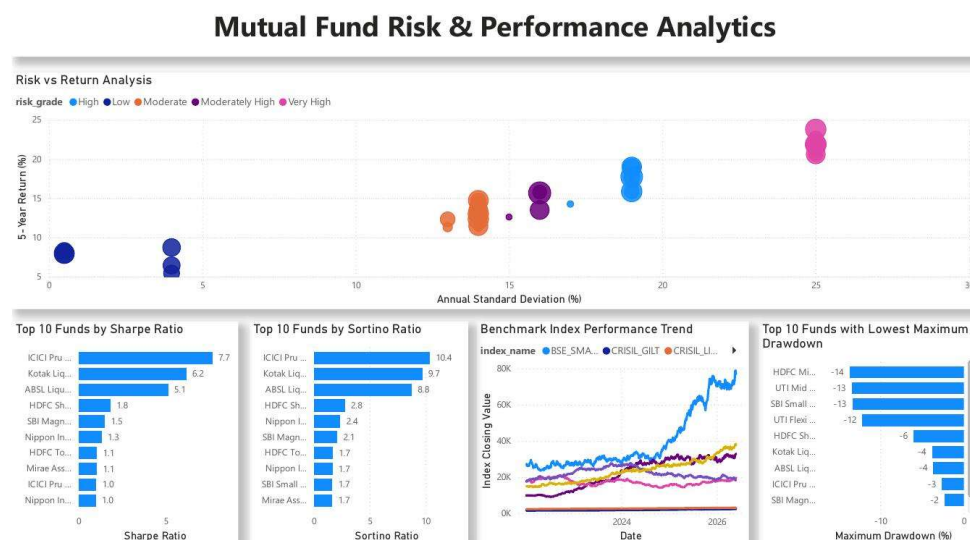


Figure 4: Mutual Fund Risk & Performance Analytics Dashboard — Power BI

This page is aimed at risk-conscious investors and portfolio analysts, pairing return data with volatility, drawdown, and risk-adjusted ratios.

- Risk vs Return Analysis: a scatter plot of 5-year return against annualized standard deviation, colour-coded by risk grade (Low, Moderate, High, Very High), visually reproducing the risk-return trade-off identified in the Python analysis.
- Top 10 Funds by Sharpe Ratio and Top 10 Funds by Sortino Ratio: both leaderboards are dominated by liquid and short-duration debt schemes (ICICI Prudential and Kotak liquid funds lead with Sharpe ratios above 6 and Sortino ratios above 9), reflecting their low volatility relative to return.
- Benchmark Index Performance Trend: closing values for indices such as BSE Small Cap, CRISIL Gilt, and CRISIL Liquid over 2024–2026, giving scheme performance a market-relative reference point.
- Top 10 Funds with Lowest Maximum Drawdown: HDFC Mid-Cap and UTI Mid Cap funds show the largest historical drawdowns (–13% to –14%) among the funds shown, while liquid and short-duration funds show minimal drawdown, underscoring the category-level risk differences visualised elsewhere on the page.

11.3 Investor Behavior & Transaction Analytics Dashboard

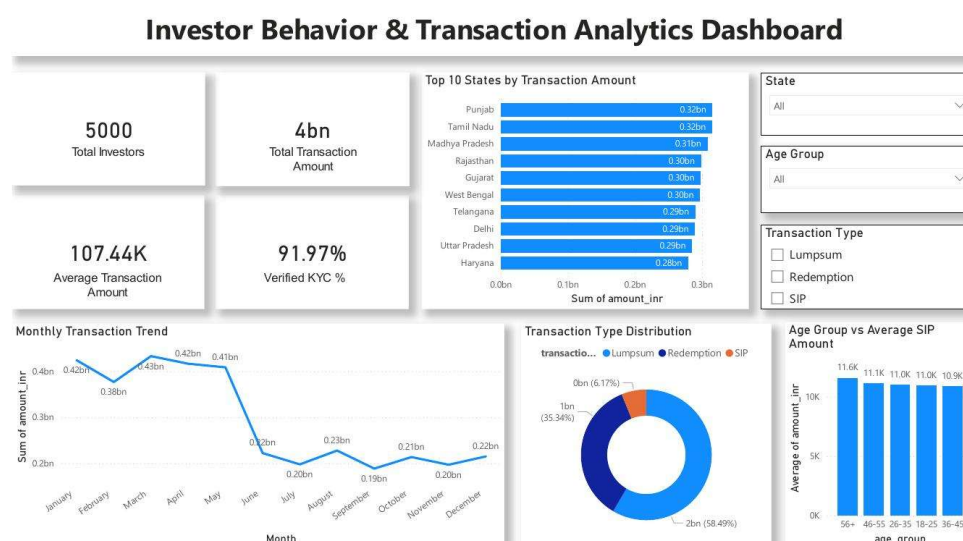


Figure 5: Investor Behavior & Transaction Analytics Dashboard — Power BI

This dashboard shifts focus from schemes to investors, summarising transaction-level behaviour across geography, age, and channel. Four KPI cards report Total Investors (5,000), Total Transaction Amount (₹4bn), Average Transaction Amount (₹107.44K), and Verified KYC % (91.97%).

- **Top 10 States by Transaction Amount:** Punjab, Tamil Nadu, and Madhya Pradesh lead the state-wise ranking, with transaction values in a tight band of ₹0.28bn–₹0.32bn across the top ten states, indicating broad geographic distribution rather than concentration in one or two states.
- **Transaction Type Distribution:** SIP transactions account for 58.49% of all transactions, Lumpsum for 35.34%, and Redemption for only 6.17%, reinforcing the EDA finding that investors strongly favour disciplined, recurring investment.
- **Age Group vs Average SIP Amount:** average SIP amounts are relatively similar across age groups (₹10.9K–₹11.6K), with the 56+ age group contributing the highest average SIP amount.
- **Monthly Transaction Trend:** transaction value fluctuates through the year, peaking in January, March, October, and November (₹0.41bn–₹0.43bn) and dipping in April and September, a pattern useful for planning seasonal marketing and service-capacity decisions.

State, Age Group, and Transaction Type slicers let AMC business and marketing teams drill into specific investor segments directly from this page.

11.4 SIP & Market Trends Dashboard

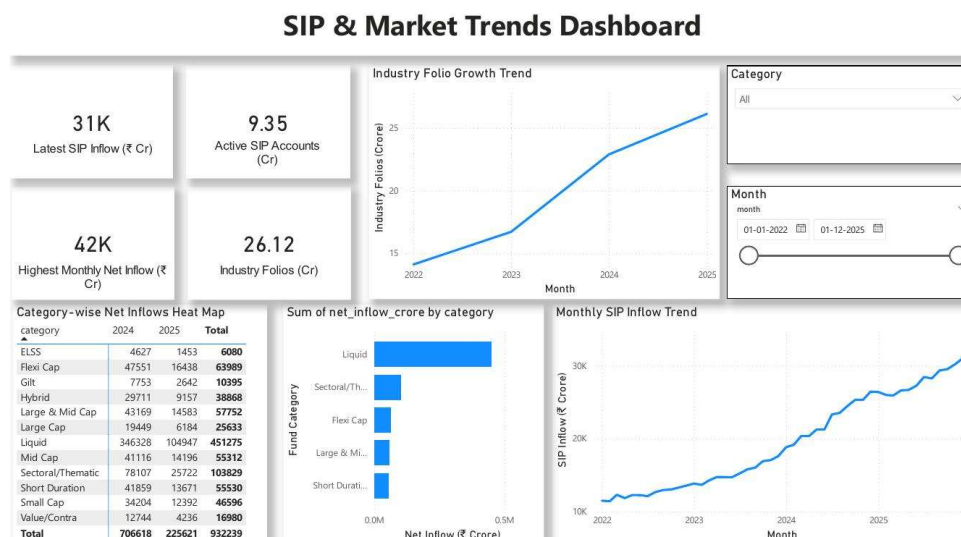


Figure 6: SIP & Market Trends Dashboard — Power BI

The final dashboard page focuses on systematic investment momentum and industry-wide folio growth over a five-year window (01-01-2022 to 01-12-2025), reporting Latest SIP Inflow (₹31K Cr), Active SIP Accounts (9.35 Cr), Highest Monthly Net Inflow (₹42K Cr), and Industry Folios (26.12 Cr) as headline KPIs.

- **Monthly SIP Inflow Trend:** reconfirms the steady rise in SIP inflows from 2022 to 2025 already seen on the industry dashboard, now paired with category-level detail.
- **Category-wise Net Inflows Heat Map:** a matrix of category versus year shows Liquid (₹451,275 Cr total), Sectoral/Thematic (₹103,829 Cr), and Flexi Cap (₹63,989 Cr) as the largest cumulative net-inflow categories across 2024–2025, with ELSS and Gilt at the lower end.
- **Industry Folio Growth Trend:** total industry folios rose steadily from 2022 through 2026, consistent with the EDA finding of a widening retail investor base.
- **Net Inflow by Fund Category (bar chart):** Liquid, Sectoral/Thematic, and Flexi Cap categories again stand out as the leading categories by net inflow volume.

A Category slicer and a Month range slider (01-01-2022 to 01-12-2025) allow this page to be used both for a full five-year retrospective and for narrower, period-specific inflow analysis.

12. Business Insights

- Equity outperformance with volatility trade-off: Small Cap and Mid Cap equity schemes consistently top the return rankings (up to 24% 5-year return) but also carry the highest volatility and largest drawdowns, confirming that return leadership in this dataset comes with materially higher risk.
- Risk-adjusted leaders differ from return leaders: liquid and short-duration debt schemes top the Sharpe and Sortino ratio leaderboards despite modest absolute returns, showing that risk-adjusted efficiency and headline return performance highlight different sets of funds — both are needed for a complete fund evaluation.
- Cost is not a reliable performance predictor: correlation and scatter analysis show no strong relationship between expense ratio and 5-year return, so expense ratio alone should not be used as a primary screening criterion for fund selection.
- SIP-led retail participation: SIP transactions make up 58.49% of all investor transactions, and SIP inflows and industry folios both show sustained multi-year growth, indicating that systematic, recurring investment — rather than lumpsum activity — is the dominant growth driver for the industry represented in this dataset.
- Concentrated fund-house and portfolio landscape: the top three fund houses (SBI, ICICI Prudential, and HDFC) account for a disproportionate share of AUM, and individual scheme portfolios are similarly concentrated in a small number of large-cap holdings and the Banking & Financial Services sector.
- Category-level demand skew: Liquid, Sectoral/Thematic, and Flexi Cap categories draw the largest net inflows, while ELSS and Gilt categories draw comparatively modest inflows — useful signal for category-level product and marketing prioritisation.

13. Recommendations

- For AMCs: continue to promote and simplify SIP onboarding, given SIP's dominant share of transactions and its consistent multi-year inflow growth; this is the clearest driver of durable, recurring AUM growth visible in the data.
- For AMCs: review portfolio and sector concentration periodically, particularly where holdings and AUM are concentrated in a small number of large-cap names and the Banking & Financial Services sector, to manage concentration risk.
- For investment advisors: present both return-based rankings and risk-adjusted rankings (Sharpe/Sortino) side by side when recommending funds, since the two rankings surface different, complementary sets of schemes.
- For investment advisors: avoid using expense ratio as a standalone fund-selection filter; combine it with return, volatility, and drawdown metrics for a more reliable recommendation.

- For retail investors: align fund selection with individual risk appetite and investment horizon — investors seeking long-term growth may consider high-return Small/Mid Cap schemes, while more conservative investors may prefer the lower-drawdown, higher-Sharpe liquid and short-duration options identified in Section 12.2.
- For marketing and product teams: use the category-wise inflow heat map (Section 12.4) to prioritise category-specific campaigns toward high-demand categories such as Flexi Cap and Sectoral/Thematic funds, while monitoring lower-demand categories like ELSS and Gilt for potential product or messaging gaps.

14. Limitations

- Dedicated data-cleaning notebook not available: the project's cleaning methodology is described in Section 7 based only on the preparation steps directly evidenced in the EDA and performance-analytics notebooks; a separate `Data_Cleaning.ipynb`, if it exists, was not part of the files supplied for this report.
- Static, point-in-time analysis: the dashboards and notebooks reflect the dataset as extracted for this project and are not connected to a live or automatically refreshing data feed; NAV, AUM, and transaction figures will require a scheduled refresh process to remain current.
- Sample-level scope: the analysis covers 40 mutual fund schemes and roughly 5,000 investor records, which is representative but materially smaller than the full universe of schemes and investors across the Indian mutual fund industry.
- No predictive modelling: the current scope covers descriptive and diagnostic analytics — what happened and why — but does not include forecasting models for future NAV, returns, or inflows.
- Metric validation limited to Sharpe and Sortino: while the Sharpe and Sortino ratios were independently recalculated and cross-checked in Python, other pre-computed fields in `scheme_performance` (such as alpha and beta) were used as supplied and were not independently re-derived in this project.

15. Conclusion & Future Scope

15.1 Conclusion

This project successfully delivered an end-to-end mutual fund analytics solution for Bluestock Fintech Pvt. Ltd., spanning relational database design in PostgreSQL, seventeen business-focused SQL queries, Python-based exploratory and performance analytics with independently validated risk metrics, and four interactive Power BI dashboards covering industry performance, risk analytics, investor behaviour, and SIP/market trends. The analysis

consistently shows that equity-oriented, higher-risk categories lead on absolute return while liquid and short-duration categories lead on risk-adjusted efficiency, that SIP-driven systematic investing is the primary engine of industry growth in this dataset, and that fund-house AUM and portfolio holdings remain concentrated among a small set of leaders. These findings, along with the accompanying recommendations, demonstrate the practical value of combining SQL, Python, and Power BI into a single analytics workflow for financial-services decision-making.

15.2 Future Scope

- Automate the ETL pipeline with a scheduled refresh from live NAV, AUM, and transaction feeds so the PostgreSQL database and Power BI dashboards stay current without manual reloads.
- Extend the performance-analytics notebook with predictive models (e.g., time-series forecasting of NAV or SIP inflow trends) to move from descriptive to predictive analytics.
- Expand the investor-behaviour analysis with cohort and retention analysis to understand SIP continuation and drop-off patterns over time.
- Incorporate additional risk metrics such as Value-at-Risk (VaR) and stress-testing scenarios to deepen the risk-analytics dashboard.
- Scale the dataset from the current sample to the full industry universe of schemes and investors as data availability permits, to improve the statistical robustness of category- and state-level insights.

16. References

- Project database schema and SQL queries: schema.sql, queries.sql (Bluestock Mutual Fund Analytics Project, Author: Saurabh Soni).
- Project Jupyter notebooks: 03_eda_analysis.ipynb, 04_performance_analytics.ipynb.
- PowerBI dashboard
file:-Mutual_Fund_Performance_Industry_Analytics_Dashboard.pbix and accompanying exported PDF.
- PostgreSQL 17 documentation
- Power BI documentation
- Pandas documentation
- Bluestock Fintech Pvt. Ltd. — Data Analyst Internship Programme guidelines.